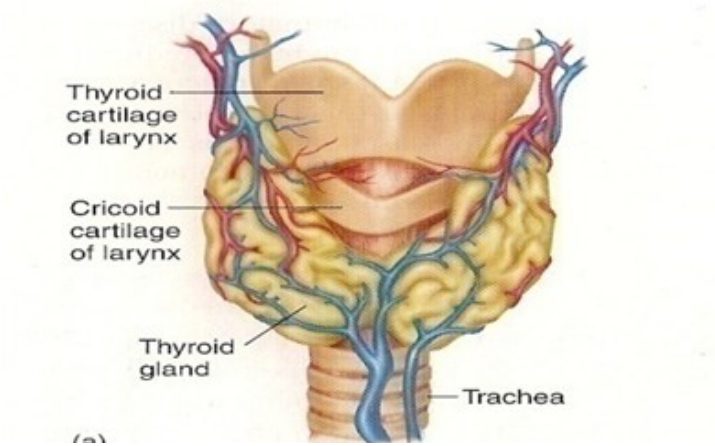
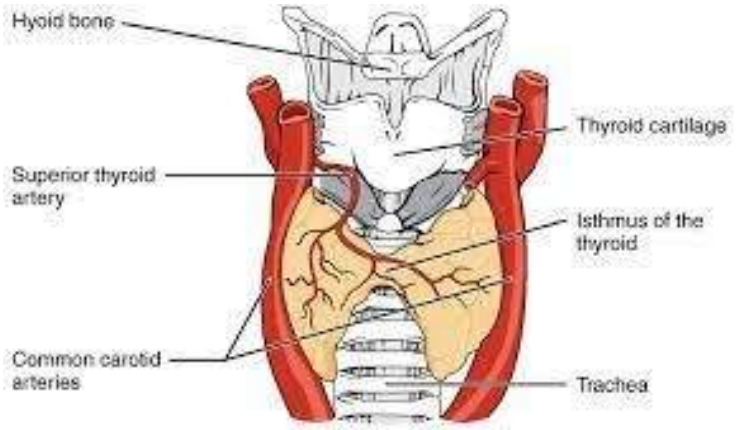
THYROID GLAND AND ITS HORMONES



Introduction

- One of largest pure endocrine glands in the body.
- Thyroid gland is well vascularized gland (rich in blood supply).
- Located in the neck just below the larynx, on either side of & anterior to the trachea.
- Formed of 2 lobes (Rt & Lt), that are connected by band of tissue called “isthmus”.



THYROID HORMONES

- **The principal hormones:**
  Thyroxine (T_4) or tetraiodothyronine
  Triiodothyronine (T_3)
  'Reverse' T3
  - Secreted by **Follicular cells.**
  - Can be stored in thyroid gland for couple of months (2-3 months).

- **Calcitonin**
  - Secreted by **Parafollicular cells.**
  - Important hormone for Ca^{2+} metabolism & homeostasis.

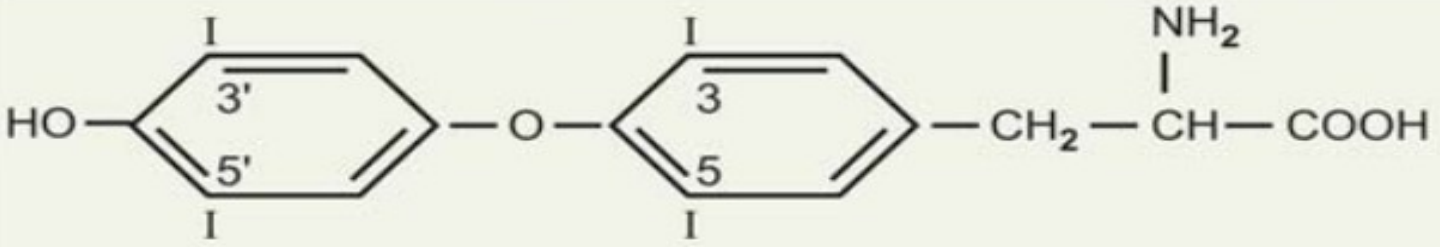
Chemistry of Thyroid Hormones:

- The hormones T4, T3 and “reverse” T3 are iodinated amino acid tyrosine.

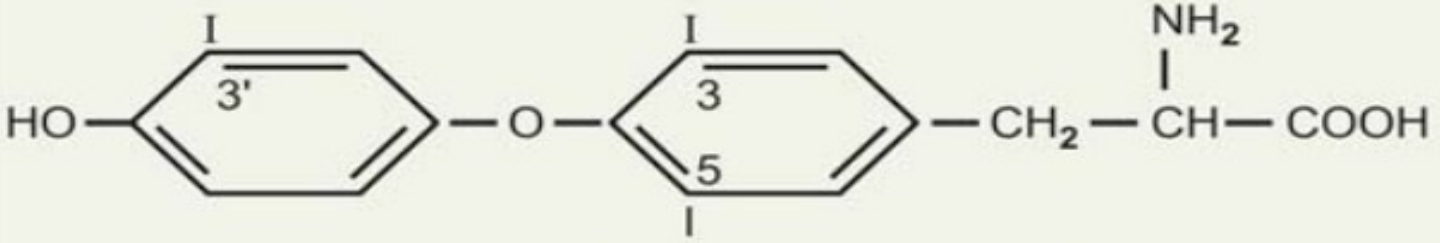
- The iodine in thyroxine accounts for 80 per cent of the organically bound iodine in thyroid venous blood.

- Small amounts of ‘reverse’ tri-iodo thyronine, Monoiodotyrosine (MIT) and other compound are also liberated.

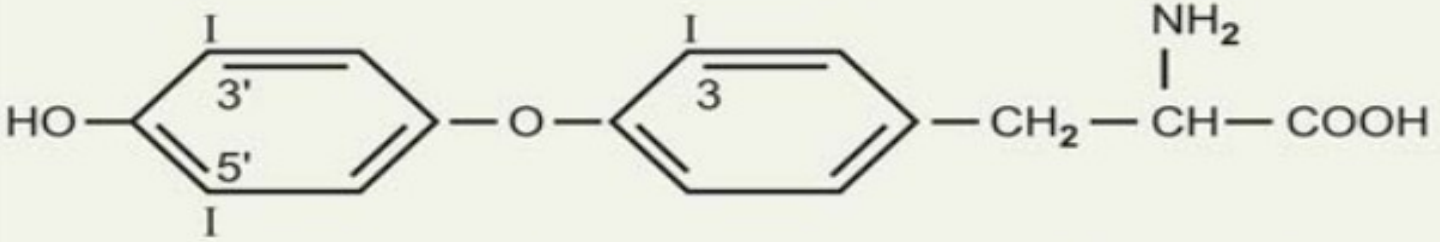
Structure of Thyroid Hormones



3,5,3',5'–tetraiodothyronine (Thyroxine, T_4)



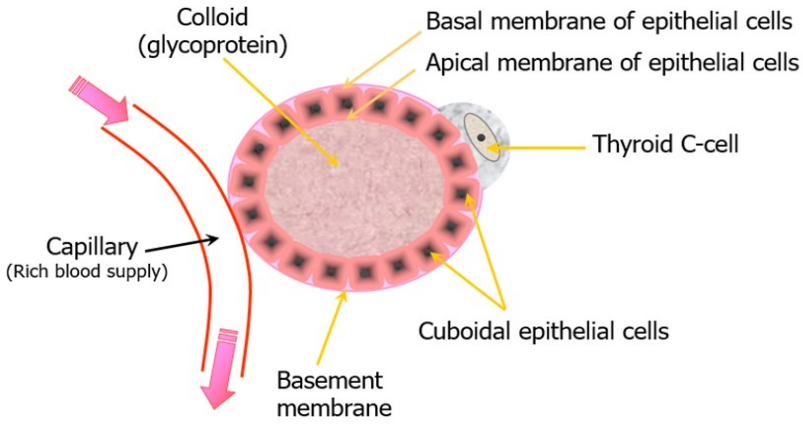
3,5,3'–Tri-iodothyronine (T_3)



3,3',5'–tri-iodothyronine ("Reverse" T_3)

BIOSYNTHESIS OF THYROID HORMONES

Structure of Euthyroid follicle –



a. Multiple Follicles (Acini):

- Are the functional unit.
- Thousands in no.
- Follicular wall is lined with a single layer of cuboidal epithelioid cells that secrete into the interior of the follicles.

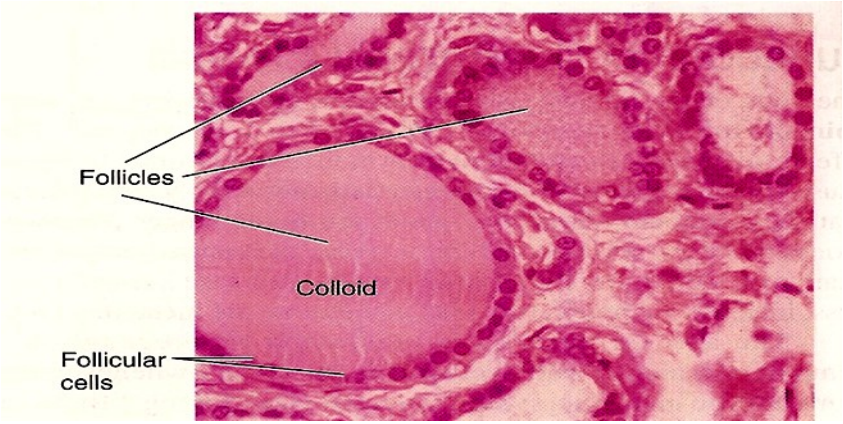
b. Parafollicular cells or "C-cells: Spherical cell, which has no relation to colloid or cuboidal cells.

- Secrete **Calcitonin**

Cont..

- When the gland is **INACTIVE**:
  colloid is abundant, follicles are large, & lining cells are flat.

- When the gland is **ACTIVE**:
  follicles are small, lining cells are cuboid or columnar



Thyroglobulin

- Each follicle is filled with proteinaceous material called colloid contains thyroglobulin.
- Thyroid hormones are synthesized by the iodination of tyrosine residue of a large glycoproteins called “thyroglobulin”.
- About 70 per cent of the iodide in thyroglobulin exists as inactive precursors ‘mono-iodo-tyrosine’ (MIT) and ‘di-iodotyrosine’ (DIT) while 30 per cent is in the iodothyronyl residues T4 and T3.
- T3 and T4 after being synthesized, remains in the bound form until it is secreted.
- When they are secreted, the peptide bonds are hydrolysed and free T3 and T4 enter the thyroid cells, cross them and are discharged into the capillaries.

Iodine Metabolism

I. The other substrate required for thyroid hormone synthesis is iodine.

II. Vegetables and fruits grown and obtained from sea-shore and also sea fishes are rich in iodine.

III. Vegetables and fruits in hilly regions lack iodine (for people residing in hilly regions, table salt should be iodinated).

IV. Ingested dietary iodine is converted to iodide and absorbed from the gut.

V. Thyroid-stimulating hormone (TSH) of anterior pituitary gland stimulates iodide uptake by the thyroid gland.

Synthesis of Thyroid Hormones

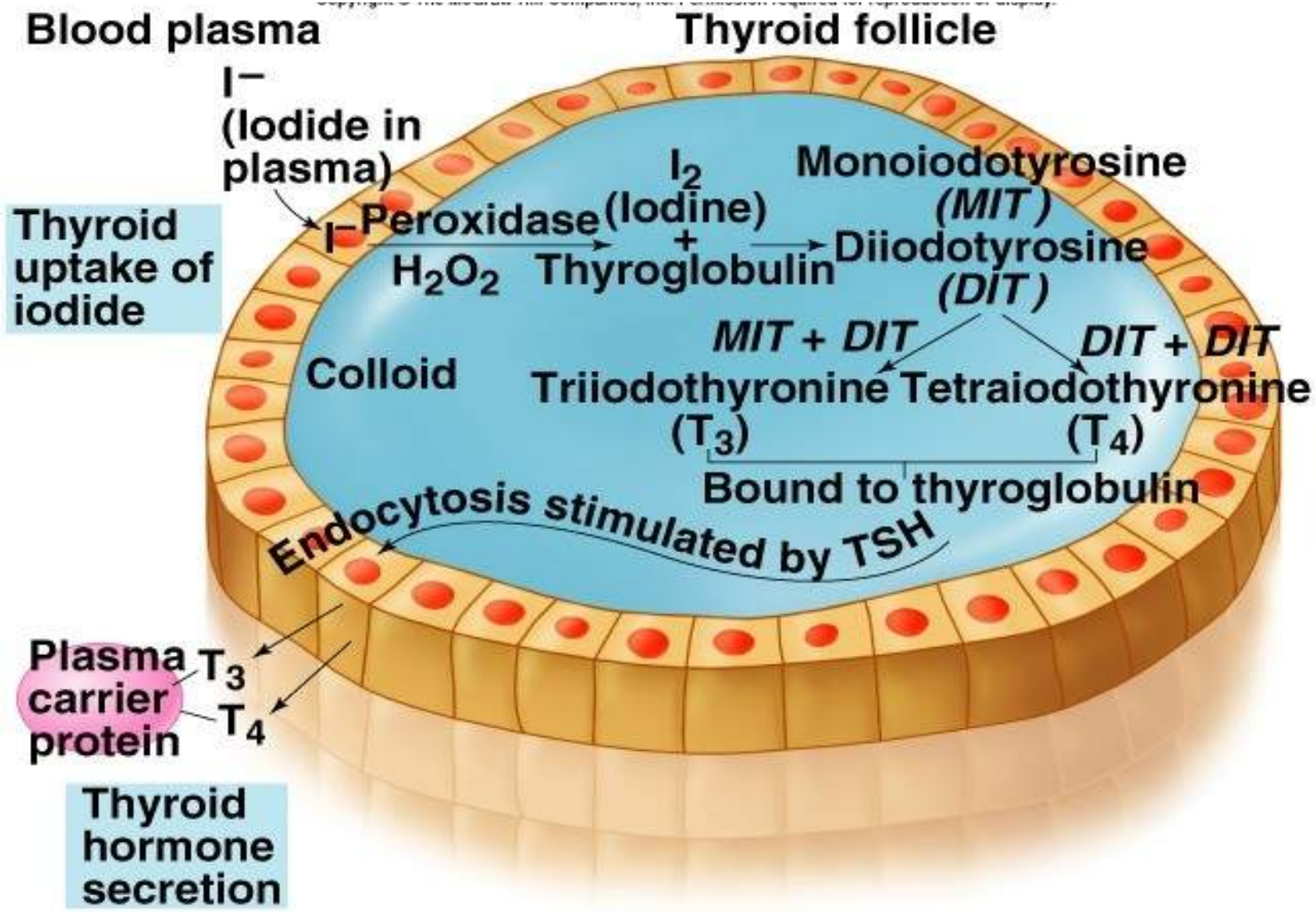
1. **Iodine trapping or "iodide-pump:** The thyroid concentrates iodide by "actively" and "selectively" transporting it from the circulation to the colloid.

2. **Oxidation of iodide:** Oxidation of iodide are catalysed by thyroperoxidase which requires H2O2 for its activity.

3. **Iodination of tyrosine:** 'Active' iodine transfers to a tyrosine residue of the thyroglobulin under the influence of thyroperoxidase enzyme.

Iodination of the tyrosine residues in thyroglobulin occurs first in "3 position" forming Mono-iodotyrosine (MIT). Monoiodotyrosine is next iodinated in the "position 5" to form "Di-iodotyrosine" (DIT).

4. **Coupling of iodotyrosines:** Two molecules of DIT when undergo an oxidative condensation forms Thyroxine (T4) molecule still in peptide linkage. Similarly, Tri-iodothyronine (T3) is probably formed by condensation of MIT with a molecule of DIT


Blood plasma
Thyroid follicle
I−
(Iodide in plasma)
Thyroid uptake of iodide
I−
Peroxidase
H2O2
I2
(Iodine)
+
Thyroglobulin
Monoiodotyrosine
(MIT)
Diiodotyrosine
(DIT)
MIT + DIT
DIT + DIT
Colloid
Triiodothyronine
(T3)
Tetraiodothyronine
(T4)
Bound to thyroglobulin
Endocytosis stimulated by TSH
Plasma carrier protein
T3
T4
Thyroid hormone secretion

Recycling of Iodine in the Gland

The hydrolysis of thyroglobulin also liberates MIT and DIT.

If these iodotyrosines are lost from the gland, considerable amounts of iodide would be biologically unavailable for the synthesis of active hormones.

Thyroid cells have microsomal enzyme de-iodinase, which uses NADPH, and they rapidly de-iodinate the iodotyrosines, and the removed iodides are recycled in the gland and utilised for thyroid hormones synthesis.

Approximately 1/3 of the total iodides in the thyroid gland is recycled in this manner.

Transport

Within the plasma, T4 and T3 are mostly transported almost entirely in association with two proteins, the so called “thyroxine-binding proteins” which act as specific carrier agents for the hormones.

Two main carrier proteins are:

- Thyroxine-binding globulin (TBG)
- Thyroxine binding prealbumin (TBPA).

Only very little T_3 & T_4 are carried in the blood in the **free active form.**

Abnormal TBG Level

In certain circumstances, TBG levels may become abnormal and it may affect binding of hormones.

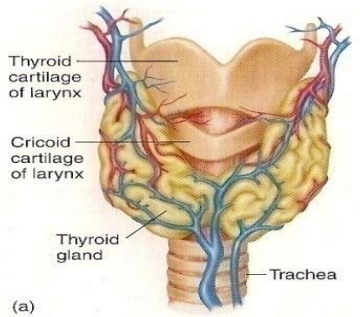
An increase occurs in:

- Pregnancy
- After administration of estrogens
- Women taking contraceptive pills.

Decrease levels seen in:

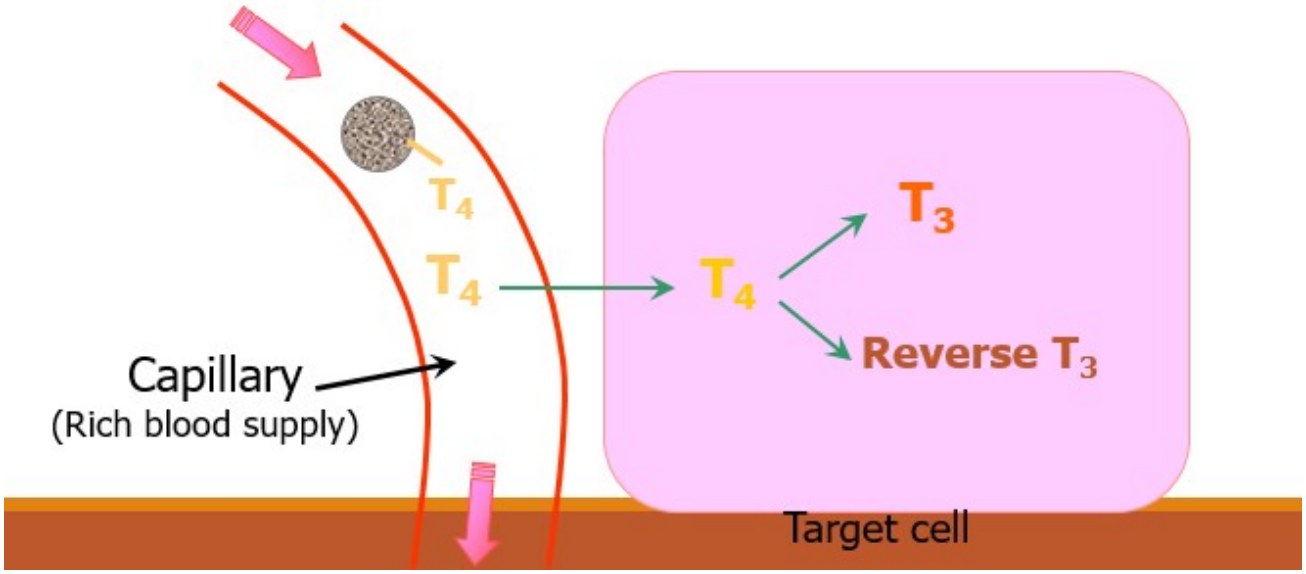
- In nephrosis
- After treatment with androgenic or anabolic steroids.
- Hypoproteinaemic states: liver diseases, cirrhosis liver

Characteristics of T3



- Extrathyroidal de-iodination converts T4 to T3
- T3 binds to the “thyroid receptor” in target cells with 10 times the affinity of T4
- About 80 per cent of circulating T4 is converted to T3 or reverse T3 (r T3) in the periphery
- T3 is loosely bound to serum proteins
- T3 is 3 to 5 times more active than T4
- Has a more rapid onset of action
- It is also more rapidly degraded in the body

T3 appears to be the major thyroid hormone metabolically

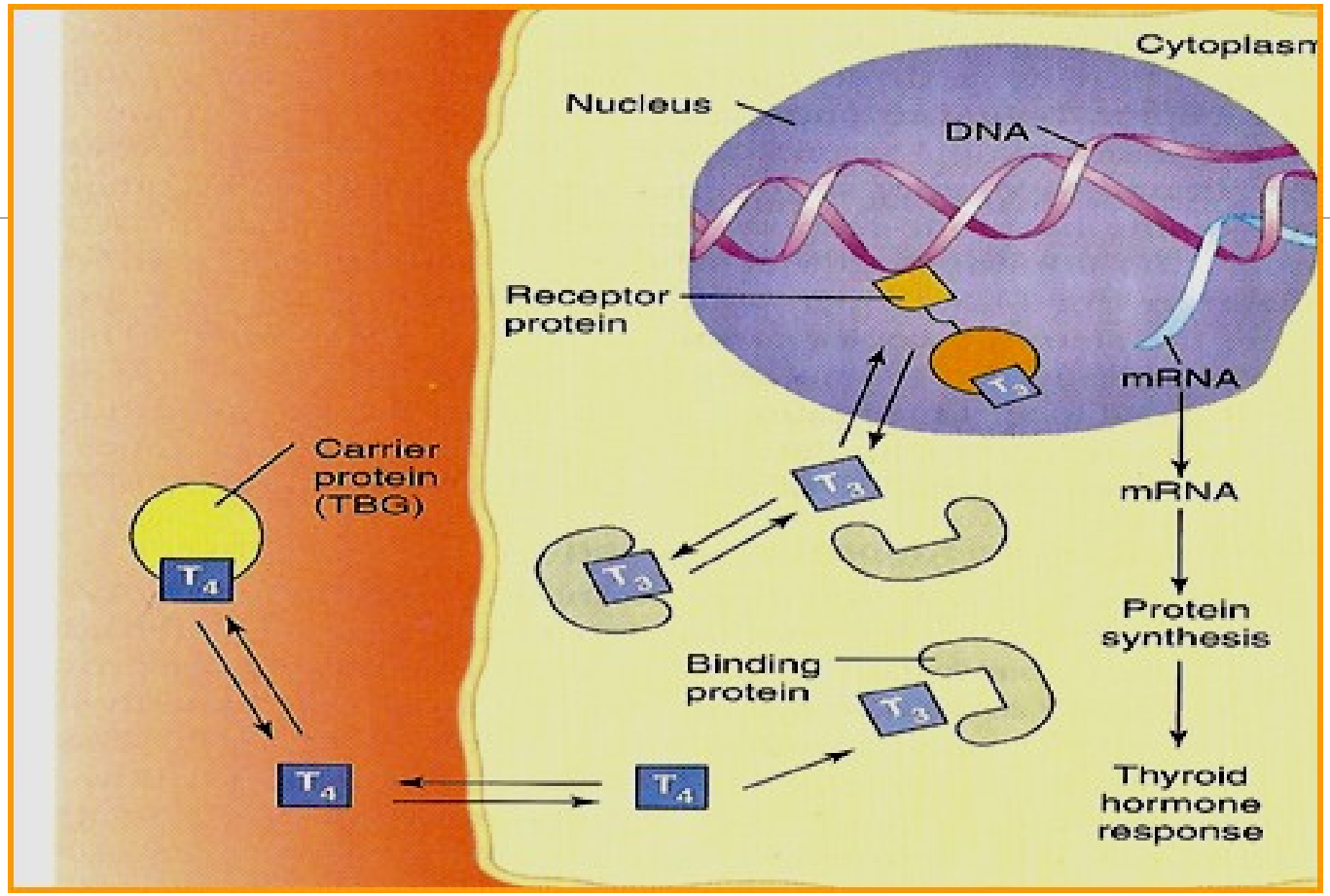


Mechanism of action of thyroid hormones

- Their receptors are located within the nucleus of target cells. This receptor hormone binding increases the action of nuclear DNA-dependent RNA polymerase increasing gene transcription, which in turn enhances m-RNA synthesis ↑ and induces synthesis of specific proteins and enzymes.

- Thyroid hormones enhance the function of Na+ K+ ATPase Pump by increasing the number of pump units, almost in all cells.

- Translation of Proteins: Besides its direct ‘nuclear action’, as stated above, thyroid hormones may stimulate translation of proteins by directly increasing the binding of amino acid–t RNA complex to ribosomes or by increasing the activity of peptidyl transferase or translocase enzymes.


Cytoplasm
Nucleus
DNA
Receptor protein
mRNA
Carrier protein (TBG)
T_4
T_3
mRNA
Protein synthesis
Binding protein
T_4
T_4
Thyroid hormone response

METABOLIC ROLE OF THYROID HORMONES

- **Generally, THs:**

1. Increases metabolic rate.
   - Stimulates increased consumption of glucose, fatty acids and other molecules.

2. Increases metabolic heat, by ↑ mitochondrial no & activity → ↑ ATP,

3. Stimulates rate of cellular respiration by:
   - Increase active transport by Na^+/K^+ pumps.
   - Stimulates O_2 consumption of most of cells in the body.

Cont..

4. Necessary for normal growth & maturation.
5. Promotes maturation of nervous system.
6. Stimulates protein synthesis.
7. Help regulating lipid & CHO metabolism.

Effects on Protein Metabolism:

• In hypothyroid children and in physiological doses, thyroid hormones when given in small doses, favour protein anabolism, leading to N-retention (+ve N-balance), because they stimulate growth.

• Large, unphysiological doses of thyroxine, cause protein catabolism, leading to –ve N-balance.

- The catabolic response in skeletal muscle, in cases of hyperthyroidism, is sometimes so severe that muscle weakness is a prominent symptom and creatinuria is marked, called thyrotoxic myopathy.

Effects on Carbohydrate Metabolism:

- Increase in blood sugar ↑ (hyperglycaemia), and glycosuria
- Increased glucose utilisation, and decreased glucose tolerance. Thyroid hormones are, therefore, antagonistic to insulin.
- Thyroid hormones increase the rate of absorption of glucose from intestine.
- Decreased glucose tolerance may be contributed to also by acceleration of degradation of insulin.
- Increased hepatic glycogenolysis ↑, because they enhance the activity of Glucose-6-phosphatase.
- In addition there is increased sensitivity to catecholamines, they potentiate the glycogenolytic effect of epinephrine.

Cont..

• Stimulate glycolysis as well as oxidative metabolism of glucose via TCA cycle and also increasing Hexose monophosphate pathway (HMP-Shunt).

• Thyroid hormones cause a decrease of glycogen store ↓ in liver and to a lesser extent, in the myocardiumand skeletal muscle.

• At the same time, thyroid hormones increase hepatic gluconeogenesis.

Effects on Lipid Metabolism

• Increases lipolysis ↑ in adipose tissue thus increasing plasma FFA ↑.

• Lipoproteins: The concentration of plasma lipoproteins (LDL) is frequently increased in hypothyroidism and decreased in thyrotoxicosis, or following administration of thyroid hormones to normal subjects.

Effects of Thyroid Hormones on the Cardiovascular System

Increase heart rate

Increase force of cardiac contractions

Increase stroke volume

Increase Cardiac output

Up-regulate catecholamine receptors

Effects of Thyroid Hormones on the Respiratory System

Increase resting respiratory rate

Increase minute ventilation

Increase ventilatory response to hypercapnia and hypoxia

Effects of Thyroid Hormones on the Renal System

Increase blood flow

Increase glomerular filtration rate

Effects of Thyroid Hormones on Oxygen-Carrying Capacity

Increase RBC mass

Increase oxygen dissociation from hemoglobin

Effects of Thyroid Hormones in Growth and Tissue Development

- Increase tooth development and eruption
- Increase growth and maturation of epidermis, hair follicles and nails
- Increase rate and force of skeletal muscle contraction
- Inhibits synthesis and increases degradation of mucopolysaccharides in subcutaneous tissue

Effects of Thyroid Hormones on the Nervous System

Critical for normal CNS neuronal development

Enhances wakefulness and alertness

Enhances memory and learning capacity

Required for normal emotional tone

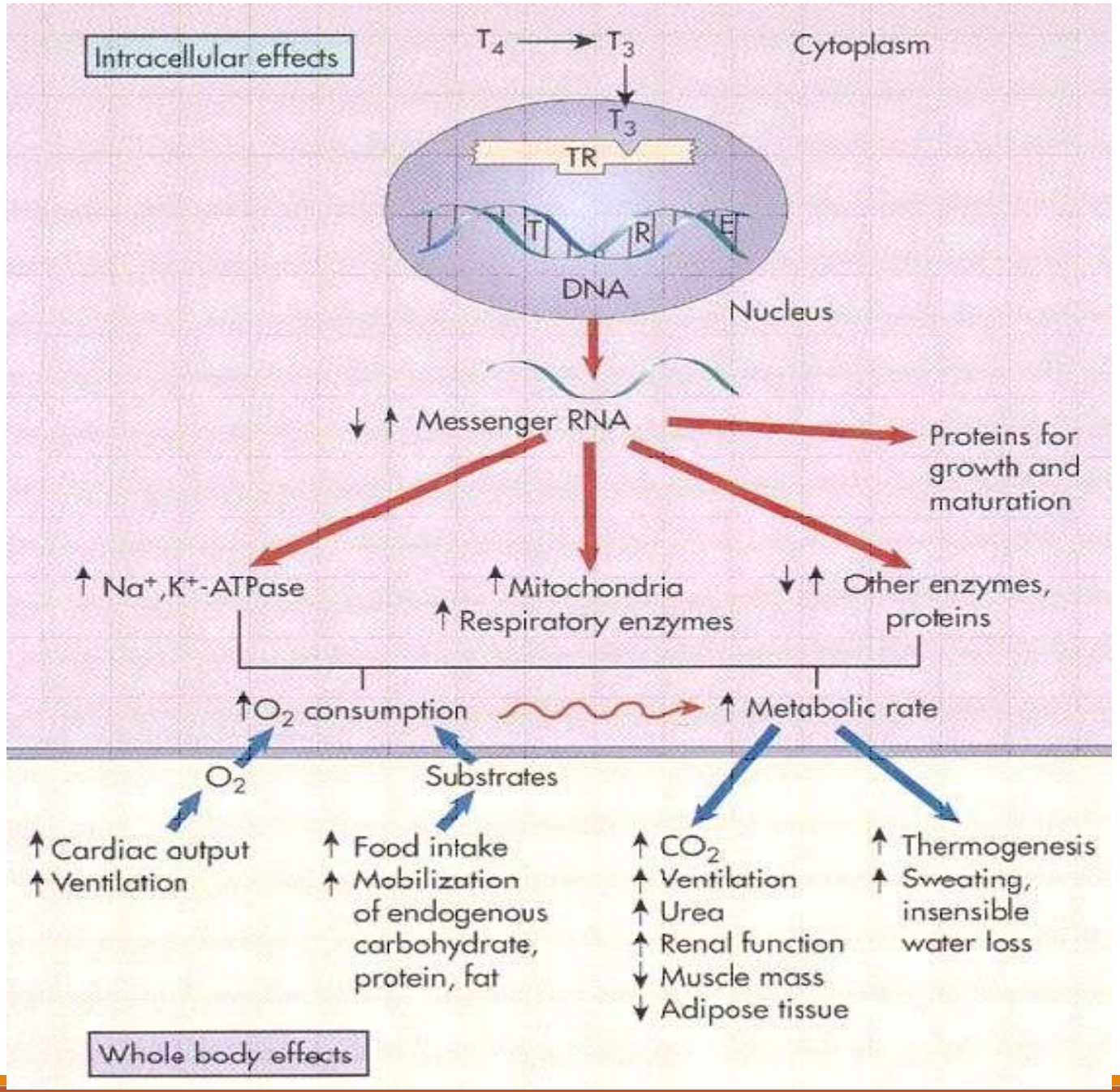
Increase speed and amplitude of peripheral nerve reflexes

Effects of Thyroid Hormones on the Reproductive System

Required for normal follicular development and ovulation in the female

Required for the normal maintenance of pregnancy

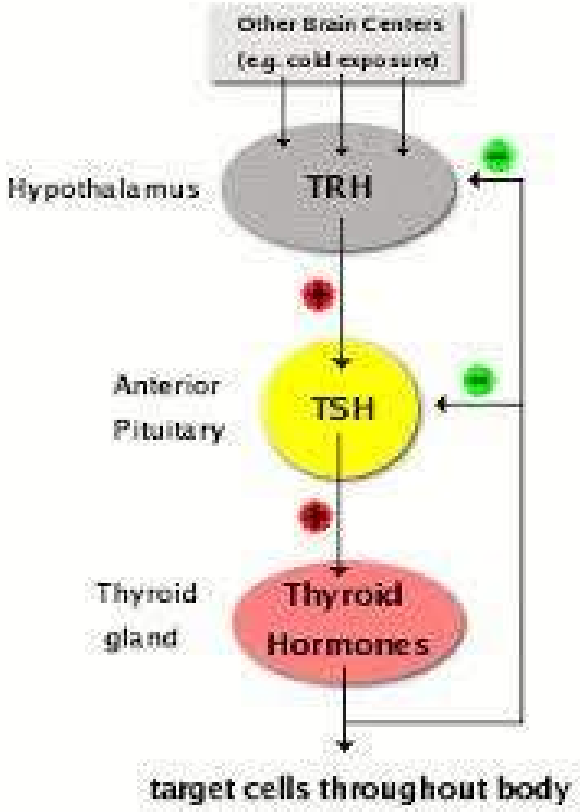
Required for normal spermatogenesis in the male


Intracellular effects
T_4 → T_3
Cytoplasm
T_3
TR
T R E
DNA
Nucleus
↓ ↑ Messenger RNA
Proteins for growth and maturation
↑ Na^+,K^+-ATPase
↑ Mitochondria
↑ Respiratory enzymes
↓ ↑ Other enzymes, proteins
↑ O_2 consumption
↑ Metabolic rate
O_2
Substrates
↑ Cardiac output
↑ Ventilation
↑ Food intake
↑ Mobilization of endogenous carbohydrate, protein, fat
↑ CO_2
↑ Ventilation
↑ Urea
↑ Renal function
↓ Muscle mass
↓ Adipose tissue
↑ Thermogenesis
↑ Sweating, insensible water loss
Whole body effects

Control of thyroid hormones secretions

- Hypothalamus (TRH) → Anterior pituitary gland (TSH) → (+) thyroid gland → (+) THs.

- THs → - ve feed back mechanism to Hypothalamus in order to inhibit (TRH) secretion to anterior pituitary gland.



Hyperthyroidism (thyrotoxicosis)

- Hyperthyoidism → ↑ THs.

- Could be:

  1ry hyperthyroidism … (diseases is in the gland), e.g. Grave's disease

  Exerts TSH-like effects on thyroid.

  Not affected by negative feedback.

  ↑ T_3 & T_4 → reflex ↓ TSH.

  2ry hyperthyroidism … (disease is higher up)

  ↑ TRH → ↑ TSH → ↑ T_3 & T_4.

- Follicular cells become overactive.
- Females > males (4:1).

Hyperthyroidism … ‘Grave’s disease’

- 90% of hyperthyoidism is due to “Grave’s disease”.
- GD is an autoimmune disease → ↑ thyroid stimulating antibodies IgG
- **Symptoms of GD:**
  - Exophthalmous, due to retro-orbital oedema (irreversible).
  - Lid lag, due to weakness of extraoccular muscles (reversible).
  - Anxiety & restlessness.
  - Sleeplessness.
  - ↑ appetite, ↓ weight & diarrhea.
  - Intolerence to heat.



Hypothyroidism
Adult (Myxedema)

- Hypothyroidism in adults → ↓ THs.
- Could be:

  1ry hypothyroidism … (diseases is in the gland)

  - autoimmune disease such as "Hashimoto's throiditis".
  - lack of iodine.
  - absence of deiodination enzyme.

  ↓ T_3 & T_4 → reflex ↑ TSH.

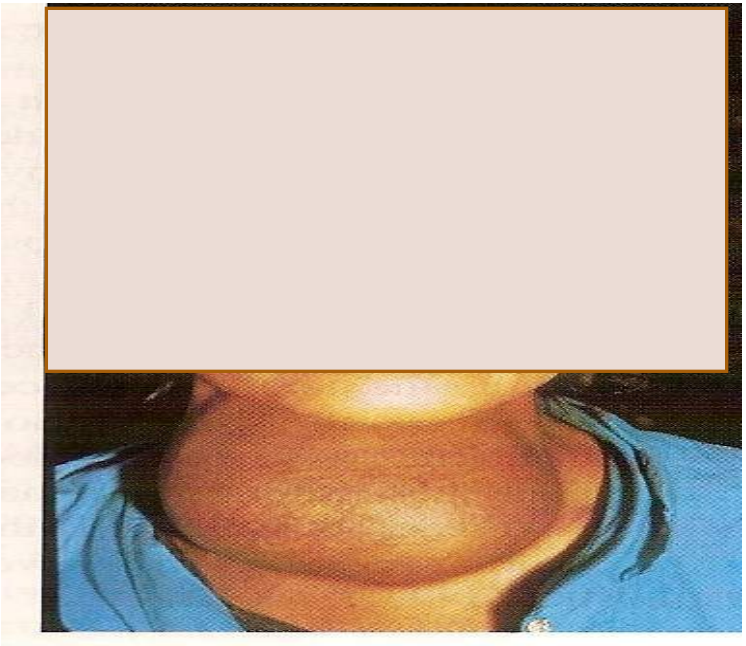
  2ry hypothyroidism … (disease is higher up)

  ↓ TRH → ↓ TSH → ↓ T_3 & T_4.

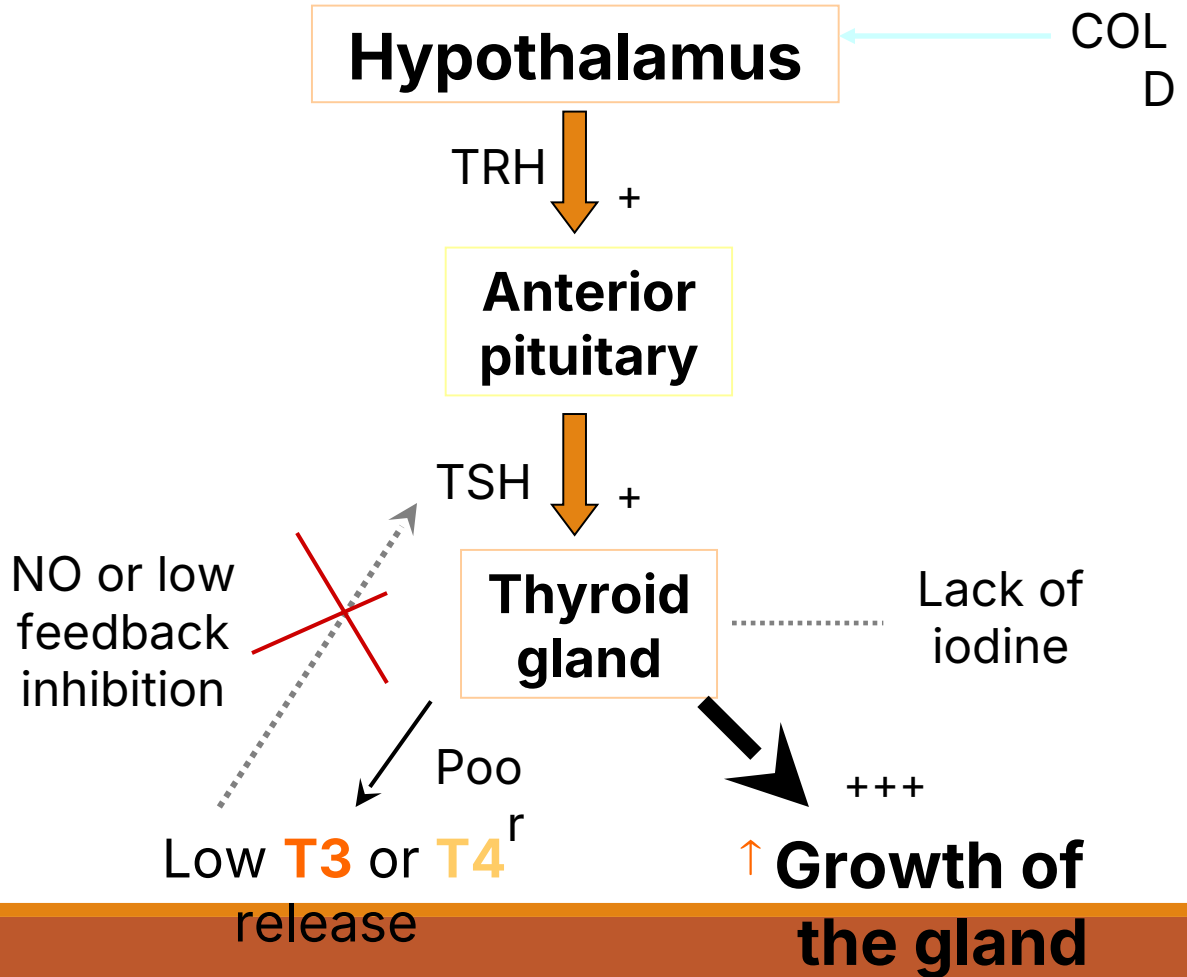
- Follicular cells become less active.

cont.

- If **No Iodine** → ↓ T_3 & T_4 → ↑ TRH → ↑ TSH→ ↑ growth (size) of the gland → **simple goiter**.



How goiter ‘swollen neck’ is formed?
With lack of iodine …



Symptoms of Hypothyroidism:

- Decreased metabolic rate.
- Slow heart rate & pulse.
- Slow muscle contractions
- ↓ appetite, ↑ weight gain, & constipation.
- Prolonged sleep, & dizziness.
- Coarse skin.
- Slow thinking, lethargy, & mask face.
- Intolerence to cold (↓ ability to adapt cold).
- Myxoedema → swollen & puffy appearance of body,
  due to deposition of protein-carbohydrate complexes
  'mucopolysaccharides' & fluid in subcutaneous tissue.

Hypothyroidism Children (Cretinism)



- Hypothyroidism in children→ ↓ THs.

  ↓ T_3 & T_4 → reflex ↑ TSH.

- **Additional Signs & Symptoms:**
  - Severe mental retardation.
  - Short stature (due to ↓ growth of bones, muscle, & brain).

Physiologic state	Serum TSH	Serum Free T4	Serum T3
Hyperthyroidism, untreated	Low	High	High
Hyperthyroidism, T3 toxicosis	Low	Normal	High
Primary Hypothyroidism, untreated	High	Low	Low or Normal
Hypothyroidism secondary to pituitary disease	Low or Normal	Low	Low or Normal

Thyroid Function Test Interpretation

TSH	Free T4	Free T3	Condition
Normal	Normal	Normal	
Low	High	High	
Low	Normal	Normal	
Low	Normal	High	
Low	High	Normal	
Low	Low	Low	
High	Normal	Normal	
High	Low	Low	
High	High	High	

CALCITONIN

- Calcitonin is a calcium regulating hormone.
- It is proved that calcitonin originates from special cells, called “C-cells”, parafollicular cells.
- C-cells constitute an endocrine system which, are derived from “neural crest” and are found in **thyroid, parathyroids and in thymus.**
- Calcitonin is a single chain polypeptide, having a mol. wt. of 3600.
- It contains 32 amino acids.

Mechanism of Action

1. **Role of Cyclic AMP:** Calcitonin binds to specific calcitonin receptors on the plasma membrane of bone osteoclasts and renal tubular epithelial cells, activates adenyl cyclase which **increases c-AMP level** which mediates the cellular effects of the hormone.

2. **Cellular Shift:** Calcitonin may directly affect the relative distribution of bone cells and number of **osteoclasts decreased.**

3. **pH Change:** Calcitonin may regulate pH at cellular level producing **more alkaline medium which diminishes resorption**.

METABOLIC ROLE

Calcitonin acts both on bone and kidneys account for:

- **Hypocalcaemia**
- **Hypophosphataemia**

Action on Bones:

- Calcitonin **inhibits the resorption of bones by osteoclasts** and thereby reduced mobilization of Ca and inorganic PO4 from bones into the blood.
- It also stimulates influx of phosphates in bones.
- There is **decrease in activities of lysosomal hydrolases, pyrophosphatase and alkaline phosphatase in bones.**
- It **increases osteoblasts cells**, which are thought to be involved in bone laying.

cont..

Action on Kidneys:

• The hormone acts on the distal tubule and ascending limb of Loop of Henle and **decreases tubular reabsorption of both calcium and inorganic PO4 t**hus producing calcinuria and phosphaturia.

• The hormone **inhibits α-1-hydroxylase and inhibits synthesis of 1,25-di-OH-D3** thus decreasing calcium absorption from intestine.

Both the above effects account for hypocalcaemia

CLINICAL ASPECT

- Abnormal calcitonin secretion is now proved in case of **“medullary carcinoma of thyroid**”.
- Histologically, medullary carcinoma arises from para-follicular C-cells of thyroid.
- Frequently patients with this tumour have been found to have associated cutaneous neuromas, adrenal tumours and parathyroid enlargement.

Therapeutic Uses of Calcitonin

Calcitonin has been used:

- In Paget's disease
- Idiopathic hypercalcaemia of infancy
- Hypercalcaemia secondary to malignancies, hyperparathyroidism and vit. D intoxication.

THANK YOU